

Chapter 3 Paper 2 — QP + MS Direct Screenshots

Updated to ignore/replace the oldest selected years: this pack uses 2017–2025 Paper 2 questions only. Both QP pages and matching MS pages are direct screenshots from repository PDFs.

Syllabus area	Covered in sections
3.1 wave motion/features/equation	1, 2, 4, 5, 8, 9
Transverse/longitudinal waves	1, 2, 3, 5, 8, 9
Ripple tank reflection/refraction/diffraction	1, 4, 5, 8, 9
3.2.1 reflection of light / plane mirror	7, 9
3.2.2 refraction, refractive index, critical angle, TIR, optical fibres-related TIR	3, 6, 8

**1) 5054/21/M/J/25 Q5 — longitudinal waves, ripple tank refraction, diffraction — QP
screenshot**

- 5 (a) Fig. 5.1 is a diagram showing the arrangement of air particles as a longitudinal wave passes through them.

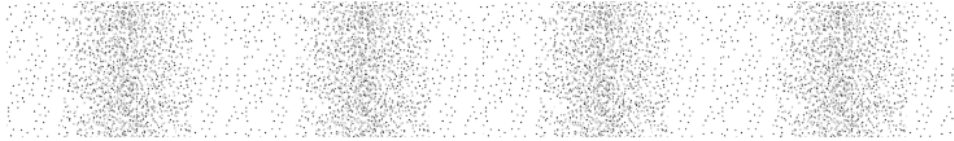


Fig. 5.1

- (i) On Fig. 5.1, mark the centre of a compression with the letter C, and mark the centre of a rarefaction with the letter R. [1]
- (ii) Describe the difference between a compression and a rarefaction.
-
-
- [1]
- (b) In a ripple tank, a water wave is produced by a wooden bar moving up and down on the surface of water.
- (i) The wooden bar makes 45 complete oscillations in 1.0 minute.
- Calculate the frequency of the wave produced.

frequency =Hz [1]

- (ii) The frequency of the water wave is increased by moving the wooden bar up and down more quickly.

State what happens to the speed and what happens to the wavelength of the wave produced.

speed

wavelength

[2]



**1) 5054/21/M/J/25 Q5 — longitudinal waves, ripple tank refraction, diffraction — QP
screenshot**



(iii) The crests of the water wave move into the shallow region shown in Fig. 5.2.

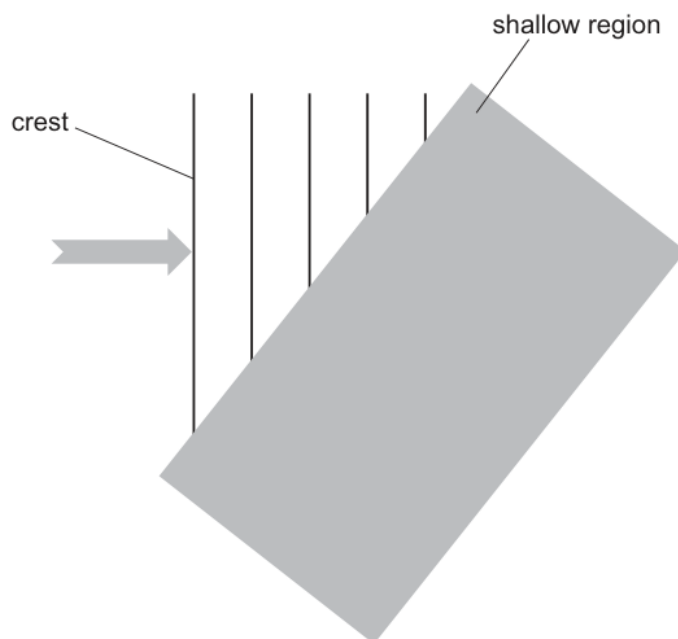


Fig. 5.2

On Fig. 5.2, draw the crests in the shallow region.

[2]

(c) Describe what is meant by the diffraction of a water wave.

.....

.....

..... [1]

[Total: 8]



1) 5054/21/M/J/25 Q5 — longitudinal waves, ripple tank refraction, diffraction — matching MS screenshot

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May/June 2025

Question	Answer	Marks
5(a)(i)	C and R marked correctly	B1
5(a)(ii)	compressions are where pressure is high / particles close together / density is high	B1
5(b)(i)	0.75 (Hz)	B1
5(b)(ii)	<i>speed</i> : no change	B1
	<i>wavelength</i> : decreases	B1
5(b)(iii)	at least 3 wavefronts in the shallow region parallel to each other and on correct side of normal	M1
	refracted towards the surface on correct side of surface and join wavefronts shown	A1
5(c)	spreading out / bending at an edge / obstacle or spreading out after passing through a (small) gap / hole	B1

2) 5054/21/O/N/24 Q5 — ultrasound, longitudinal vs transverse, wavelength — QP
screenshot



- 5 Ultrasound is a longitudinal wave which cannot be heard by humans as the frequency of ultrasound is too high.

- (a) Ultrasound waves **cannot** travel in a vacuum.

Explain why.

.....
.....
..... [2]

- (b) Describe how a longitudinal wave differs from a transverse wave.

.....
.....
.....
..... [2]

- (c) An ultrasound transmitter used for a medical scan produces an ultrasound wave of frequency 8.4 MHz.

In soft human tissue, ultrasound travels at 1500 m/s.

- (i) Calculate the wavelength of this ultrasound wave in soft human tissue.

wavelength = m [3]

- (ii) The ultrasound wave passes from the soft tissue into bone where the speed of the ultrasound wave is greater than 1500 m/s.

State what happens to the frequency and to the wavelength of the ultrasound wave as it passes into the bone.

frequency

wavelength

[2]

[Total: 9]



2) 5054/21/O/N/24 Q5 — ultrasound, longitudinal vs transverse, wavelength — matching MS screenshot

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PUBLISHED

October/November 2024

Question	Answer	Marks
4(c)(ii)	frequency increases and wavelength decreases	B1
4(d)	infrared (radiation)	B1

Question	Answer	Marks
5(a)	ultrasound / sound / it is transmitted by <u>vibrating</u> matter / particles	B1
	no matter / particles in a vacuum / no medium	B1
5(b)	(in a longitudinal wave,) vibration (of particles) is parallel to the propagation direction	B1
	in a transverse wave, vibration (of particles) is perpendicular to the propagation direction	B1
5(c)(i)	$(\lambda =) v / f$ or $1500 / 8.4 \times 10^6$	C1
	1.8×10^N	C1
	1.8×10^{-4} (m)	A1
5(c)(ii)	frequency: does not change	B1
	wavelength: increases	B1

3) 5054/21/O/N/23 Q4-Q5 — light refraction + transverse/longitudinal comparison — QP screenshot

- 4 Red light of frequency 4.7×10^{14} Hz travels in air at a speed of 3.0×10^8 m/s.

(a) Calculate the wavelength in air of this red light.

wavelength = m [3]

- (b) Fig. 4.1 shows a narrow beam of this red light striking the surface of a parallel-sided glass block.

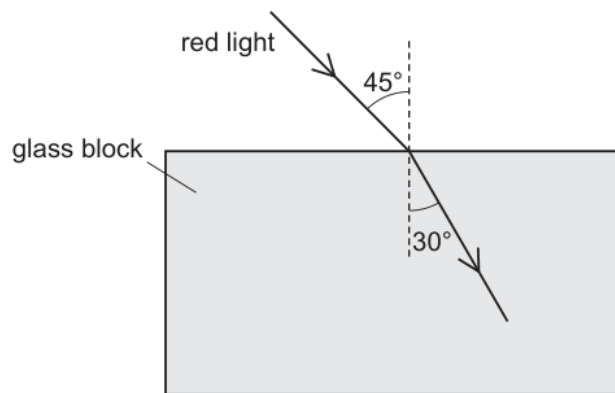


Fig. 4.1

- (i) State what happens to the speed, to the frequency and to the wavelength of the light as it enters the block.

speed

frequency

wavelength

[2]

- (ii) Using the angles shown on Fig. 4.1, calculate the refractive index of the glass.

refractive index = [3]

3) 5054/21/O/N/23 Q4-Q5 — light refraction + transverse/longitudinal comparison — QP screenshot

- (iii) The light continues along the path shown in Fig. 4.1 until it strikes the bottom surface of the block. Light then emerges into the air.

Draw on Fig. 4.1 to show the path taken by the light until it strikes the bottom surface and the path of the light that emerges into the air. [2]

[Total: 10]

3) 5054/21/O/N/23 Q4-Q5 — light refraction + transverse/longitudinal comparison — QP screenshot

5 Most of the energy emitted by the Sun is in three regions of the electromagnetic spectrum.

(a) The name of one of these three regions is the ultraviolet region.

(i) State the names of all three regions in order of increasing frequency.

smallest frequency

intermediate frequency

largest frequency [2]

(ii) Some of the ultraviolet radiation emitted by the Sun reaches the surface of the Earth.

Give one possible damaging effect of ultraviolet radiation on the human body and the property of ultraviolet radiation that causes this damage.

damaging effect

.....

property

..... [2]

(b) Electromagnetic radiation is a transverse wave.

Describe the difference between a transverse wave and a longitudinal wave.

.....

.....

..... [2]

[Total: 6]

3) 5054/21/O/N/23 Q4-Q5 — light refraction + transverse/longitudinal comparison — matching MS screenshot

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Question	Answer	Marks
4(a)	$(\lambda =) v/f$ or $3.0 \times 10^8 / 4.7 \times 10^{14}$	C1
	6.4×10^{-4} (m) or $3 / 4.7 \times 10^6$	C1
	6.4×10^{-7} (m)	A1
4(b)(i)	speed: decreases frequency: stays the same wavelength: decreases	B2
4(b)(ii)	$i = 45^\circ$ and $r = 30^\circ$	C1
	$(n =) \sin(i) / \sin(r)$ or $\sin(45^\circ) / \sin(30^\circ)$	C1
	1.4	A1
4(b)(iii)	straight continuation of ray in block and refraction away from normal in air	B1
	emergent ray parallel to initial incident ray	B1

Question	Answer	Marks
5(a)(i)	infrared (radiation) and visible (light) and ultraviolet (radiation)	B1
	infrared (radiation) and visible (light) and ultraviolet (radiation) and in this order	B1
5(a)(ii)	damaging effect: (skin) cancer / cataracts	B1
	property: it / ultraviolet radiation is ionising	B1

3) 5054/21/O/N/23 Q4-Q5 — light refraction + transverse/longitudinal comparison — matching MS screenshot

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October/November 2023

Question	Answer	Marks
5(b)	in a transverse wave, the <u>vibration</u> direction is perpendicular / at right angles to the propagation direction	B1
	in a longitudinal wave, the <u>vibration</u> direction is parallel to the propagation direction	B1

Question	Answer			Marks
6(a)	(because) the current at every point in a series circuit is the same			B1
6(b)(i)	$(R =) V / I$ or $1.5 / 0.20$ or $7.5 (\Omega)$			C1
	$(R_T =) 7.5 - 1.5 - 3.5$			C1
	2.5 (Ω)			A1
6(b)(ii)	resistance (of T / thermistor) decreases (as the temperature increases)			B1
	current in circuit increases	or	smaller fraction of total resistance	B1
	p.d. across R and S increases	or	(p.d. is) smaller fraction of e.m.f.	B1

Question	Answer	Marks
7(a)(i)	P: (carbon) brush and Q: slip ring(s)	B1
7(a)(ii)	to connect the output terminal(s) and the (rotating) coil/brushes	B1
	without wires twisting	B1
7(b)(i)	at least one of the five positions when the output e.m.f. is zero indicated with X and no indications elsewhere	B1

**4) 5054/21/M/J/23 Q7 — diffraction through a gap + wavelength definition — QP
screenshot**

- 7 A water wave in a ripple tank diffracts as it passes through the gap in a barrier.

Fig. 7.1 shows a drawing made by a student of the crests in the pattern.

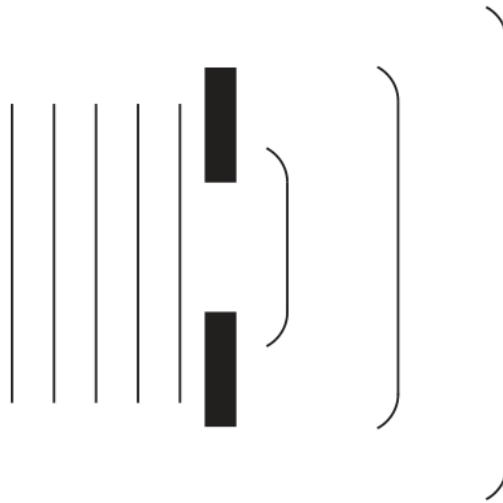


Fig. 7.1

- (a) State **one** way in which the student's drawing of the crests is wrong.

.....
 [1]

- (b) The gap in the barrier is now made smaller than the wavelength, as shown in Fig. 7.2.

Complete Fig. 7.2 with at least three crests to show the new diffraction pattern.

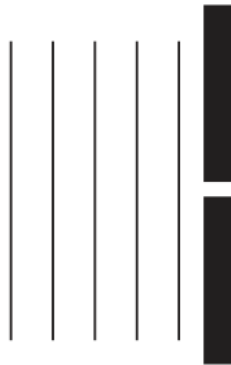


Fig. 7.2

[2]

4) 5054/21/M/J/23 Q7 — diffraction through a gap + wavelength definition — matching MS screenshot

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May/June 2023

Question	Answer	Marks
7(a)	wavelength has altered / too large or crests in shadow should be bent / quarter circular above or below gap or vertical lines too straight on two crests	B1
7(b)	at least 3 semicircles centered on the gap	B1
	wavelength constant and the same on right and left of gap and some bending	B1
7(c)(i)	distance between two (consecutive) identical points such as two (consecutive) crests	B1
7(c)(ii)	$v = f\lambda$ numerical or algebraic e.g. $f = 6 / 2$ (Hz)	C1
	3(.0) Hz)	A1
7(c)(iii)	1.3 (cm)	B1

**5) 5054/22/M/J/23 Q5 — transverse examples and diffraction at an edge — QP
screenshot**

5 Water waves are transverse waves.

(a) Underline **two** other examples of transverse waves.

seismic P-waves

seismic S-waves

sound

X-rays

[1]

(b) Fig. 5.1 shows a wooden bar and a glass block in a ripple tank. The depth of water in the tank is less than the height of the glass block.

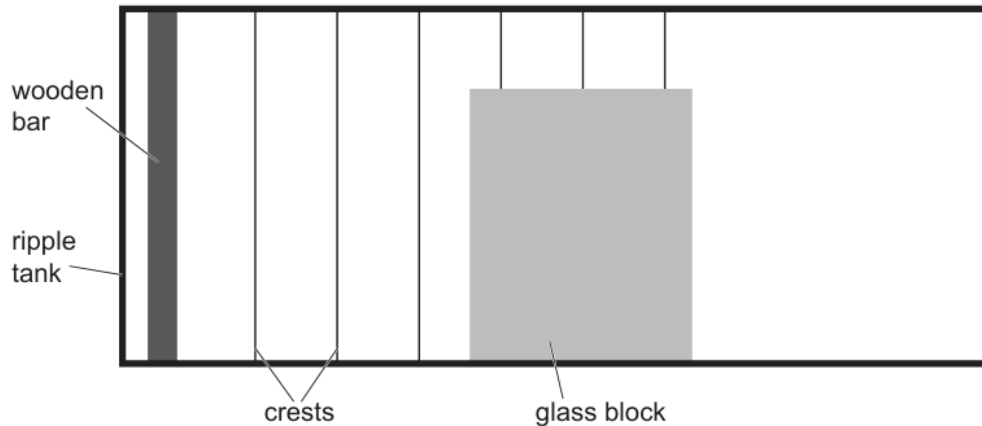


Fig. 5.1 (not to scale)

The wooden bar moves up and down once every 0.15 s to create the crests.

(i) The speed of the water wave is 27 cm/s.

Calculate the frequency and the wavelength of the wave.

frequency = Hz

wavelength = cm
[3]

(ii) The wave diffracts at the right-hand edge of the glass block.

On Fig. 5.1 draw **two** crests after they pass the glass block to show the diffraction. [2]

(iii) Describe how a wave with a smaller wavelength is made with the wooden bar.

.....
..... [1]

(iv) Describe how a decrease in wavelength affects the diffraction.

.....
..... [1]

5) 5054/22/M/J/23 Q5 — transverse examples and diffraction at an edge — matching MS screenshot

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May/June 2023

Question	Answer	Marks
4(a)(i)	–273 (°C) and 0 (K)	B1
4(a)(ii)	particles) move fast(er)	B1
	<u>vibrate</u> / <u>oscillate</u> with larger amplitude	B1
4(a)(iii)	internal energy increases	B1
	temperature stays the same	B1
4(b)(i)	($E = mcT$) in any form numerical or algebraic	C1
	8000 (J)	A1
4(b)(ii)	100–44 or 56	C1
	0.48 or 0.47 (J/(g °C))	A1

Question	Answer	Marks
5(a)	seismic S-waves and X-rays	B1
5(b)(i)	(wavelength =) v / f in any form or (distance =) vt or $f = 1 / T$ or $1 / 0.15$	C1
	frequency 6.7 (Hz)	A1
	wavelength 4.0 (cm)	A1
5(b)(ii)	<u>straight</u> wavefronts above block to the right with same wavelength	B1
	correct diffraction – wavefronts quarter circles with centre at top corner	B1
5(b)(iii)	increase frequency of movement of (wooden) bar	B1

5) 5054/22/M/J/23 Q5 — transverse examples and diffraction at an edge — matching MS screenshot

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PUBLISHED

May/June 2023

Question	Answer	Marks
5(b)(iv)	less diffraction or less bending	B1

Question	Answer	Marks
6(a)(i)	ammeter A_2 and A_3 both 0.25 (A)	B1
	voltmeter V_2 9(.0 V)	B1
6(a)(ii)	($R =$) V / I in any form	C1
	36 (Ω)	A1
6(a)(iii)	current is (directly) proportional to voltage / potential difference	B1
	provided the temperature is constant	B1
6(b)	0.35 (A) seen or total resistance $36 + 6 / 0.35$; $36 + 17(.1)$; $53(.1) \Omega$	B1
	voltage (across R) 0.35×36 or 12.6 (V) seen	B1
	19 V	B1

Question	Answer	Marks
7(a)(i)	X-rays and gamma (radiation / rays)	B1
7(a)(ii)	security marking / detecting counterfeit bank notes / sterilising water / fluorescent effects (e.g. fingerprints, blood, biological fluids) / (sun) tanning / sun beds / cancer treatment / curing inks and resins (e.g. in dental fillings) / production of vitamin (D) / generating electricity / solar panels / disinfection / sterilisation	B1
7(a)(iii)	(skin) cancer / cataracts / burns or destroys the retina or cornea	B1

**6) 5054/22/O/N/21 Q3 — refraction, refractive index, critical angle/TIR — QP
screenshot**

- 3 Fig. 3.1 shows light entering a transparent block.

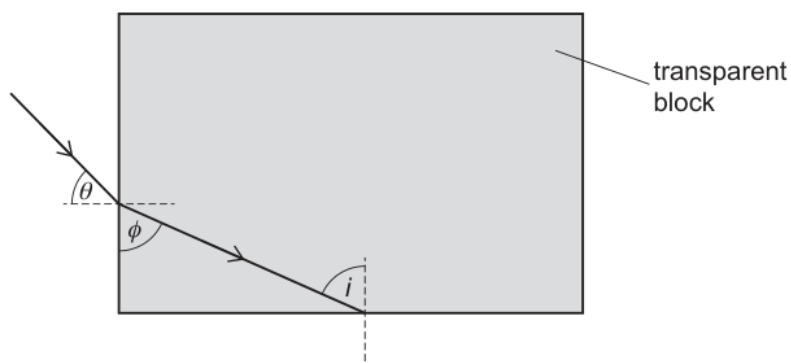


Fig. 3.1 (not to scale)

The light enters the block at an angle θ to the normal and travels through the block until it meets the bottom surface.

The angle between the ray in the block and the vertical side of the block is ϕ .

- (a) Light travels more slowly in the block than in air.

- (i) Explain how Fig. 3.1 shows this.

.....

 [2]

- (ii) State what happens to the wavelength of the light and what happens to the frequency of the light as it enters the block.

wavelength
 frequency
 [2]

- (b) The refractive index of the transparent material is 1.6. Angle θ is 45° .

- (i) Determine angle ϕ .

$\phi =$ [3]

**6) 5054/22/O/N/21 Q3 — refraction, refractive index, critical angle/TIR — QP
screenshot**

- (ii) The angle of incidence i at the bottom surface is equal to ϕ and the critical angle for the material of the block in air is 39° .

Explain what happens to the light after it meets the bottom surface.

.....

.....

..... [2]

[Total: 9]

6) 5054/22/O/N/21 Q3 — refraction, refractive index, critical angle/TIR — matching MS screenshot

5054/22

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PUBLISHED

October/November 2021

Question	Answer	Marks
2(b)(ii)	(water) resistance / resistive force / friction (on swimmer)	B1
	(water) resistance / backward force / resistive force / friction / opposing force increases (with speed / time)	B1
	(at constant speed) forward force is equal to resistive / backward force or no resultant force or forces balance	B1

Question	Answer	Marks
3(a)(i)	it / light refracts / bends / moves towards normal or angle of incidence greater than angle of refraction	B1
	change in direction of the ray / refraction shows a change in speed (at the boundary) or $\sin(i) / \sin(r)$ is ratio of speeds or top of ray slows down first and enters the block first	B1
3(a)(ii)	(wavelength) decreases	B1
	(frequency) does not change.	B1
3(b)(i)	$(\sin(r) = \sin(i) / n$ or $\sin(45^\circ) / 1.6$ or $0.71 / 1.6$ or 0.44 or $(r = 21^\circ)$	C1
	26°	C1
	64°	A1
3(b)(ii)	64° / angle of incidence / i is greater than the critical angle / 39° or angle of incidence = 64°	B1
	it undergoes total internal reflection	B1

Question	Answer	Marks
4(a)(i)	13 A	B1
4(a)(ii)	0 (A)	B1

6 A dentist uses a plane mirror to see the back of a tooth.

(a) A plane mirror produces an image of an object.

Describe the position of this image.

.....
.....
..... [2]

(b) Fig. 6.1 shows the plane mirror used by the dentist to see the point labelled X on the tooth.

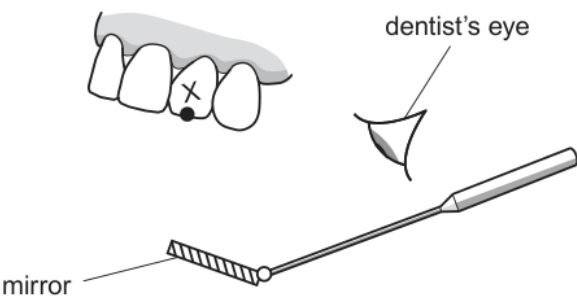


Fig. 6.1

On Fig. 6.1:

- (i) mark the position of the image of X formed by the mirror [1]
 - (ii) draw a ray of light from X to show how the dentist can see the tooth. [2]
- (c) State **one** characteristic of the image formed by the plane mirror other than its position.

.....
..... [1]

[Total: 6]

7) 5054/21/M/J/20 Q6 — plane mirror image and ray construction — matching MS screenshot

5054/21

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PUBLISHED

May/June 2020

Question	Answer	Marks
5(a)	pressure = force / area	B1
	sharp knife has small(er) area and large(r) pressure or blunt knife has large(r) area and small(er) pressure	B1
5(b)(i)	temperature	B1
	mass of gas or number of molecules	B1
5(b)(ii)	molecules hit sides of container	B1
	twice as many molecules per second hit (unit area) or molecules hit twice as frequently	B1

Question	Answer	Marks
6(a)	behind the mirror or on the opposite side (of mirror) to object	B1
	same distance as object (from mirror)	B1
6(b)(i)	image marked at correct position by eye	B1
6(b)(ii)	ray from tooth to mirror and ray from mirror to eye	M1
	angles of incidence and reflection equal (by eye)	A1
6(c)	virtual or same size (as object) or laterally inverted	B1

9 Some waves are longitudinal and some waves are transverse.

(a) State how a longitudinal wave differs from a transverse wave.

.....
..... [1]

(b) A vibrating rod produces a water wave in a ripple tank. Fig. 9.1 shows the crests of the wave passing into the right-hand section of the tank where the depth of the water is different from the depth in the rest of the tank.

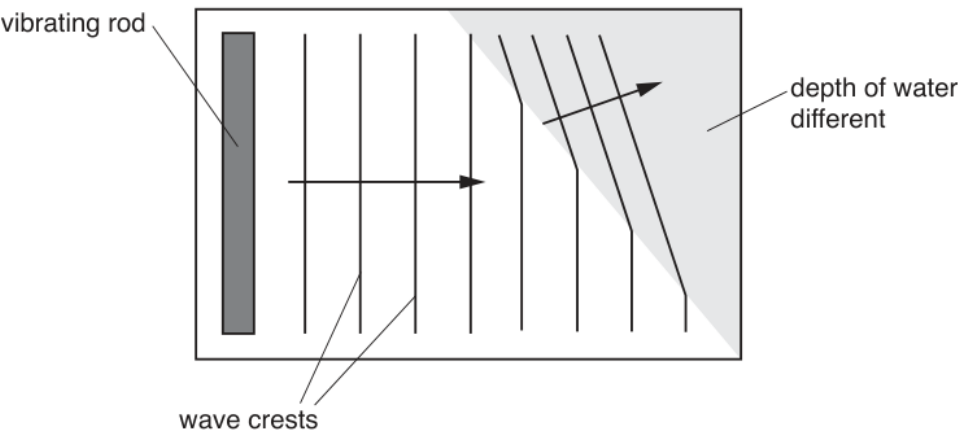


Fig. 9.1

The arrows on Fig. 9.1 show the direction of travel of the wave in the two sections of the ripple tank.

(i) In the left-hand section of the tank, the wavelength of the wave is 0.019 m and it travels at 0.17 m/s.

Calculate the frequency of the wave.

frequency = [2]

(ii) State what happens to the frequency of the wave, as it passes into the right-hand section of the tank.

..... [1]

(iii) Using Fig. 9.1, state and explain what happens to the speed of the wave as it passes into the right-hand section of the tank.

.....
.....
..... [2]

(c) Fig. 9.2 shows light, in air, striking the vertical side of a rectangular glass block at an angle of incidence of 60° .

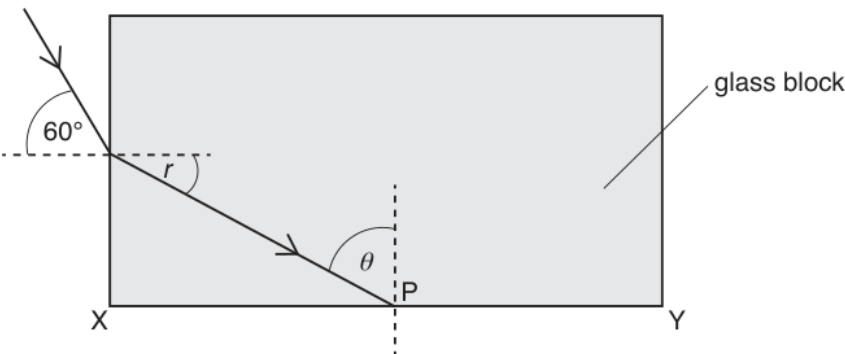


Fig. 9.2

The refractive index of the glass is 1.6. The light travels in the glass and strikes side XY at P.

(i) Underline all the terms that describe a light wave.

electromagnetic **longitudinal** **transverse**

[1]

(ii) At the point where the light enters the glass, the angle of refraction is r .

Calculate angle r .

$r =$ [2]

(iii) 1. Calculate the critical angle c for light travelling in the block.

$c =$ [2]

2. At P, the angle θ between the ray and the normal is given by $\theta = 90^\circ - r$.

State and explain what happens to the light when it strikes side XY.

.....
.....
..... [2]

3. On Fig. 9.2, draw the path of the light after it strikes side XY at P and the path of the light when it is again travelling in the air. [2]

8) 5054/21/O/N/18 Q9 — ripple-tank refraction and light refraction — matching MS screenshot

5054/21

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PUBLISHED

October/November 2018

Question	Answer	Marks
8(d)(ii)	from chemical (energy) or chemical (energy) as first term	B1
	to gravitational potential (energy) or to thermal / heat (energy) as last term	B1
	to thermal / internal (energy) / heat and thermal / internal (energy) / heat	B1
8(e)	no resultant force or forces balance / cancel or tension equals weight (of bucket) or upwards force equals downward force	B1

Question	Answer	Marks
9(a)	the <u>vibration</u> direction is parallel to the wave / energy travel direction or longitudinal waves have compressions and rarefactions or transverse waves have crests and troughs or longitudinal waves cannot be polarised	B1
9(b)(i)	$(f =) v / \lambda$ or 0.17 / 0.019	C1
	8.9 Hz	A1
9(b)(ii)	(frequency) stays constant or does not change	B1
9(b)(iii)	(speed) decreases	B1
	(Fig. 9.1 shows that) wavelength decreases or refraction is towards normal or $i > r$ or top of wave (in shaded region) lags behind bottom of wave	B1
9(c)(i)	electromagnetic and transverse underlined	B1
9(c)(ii)	$n = \sin(i) / \sin(r)$ or $(r =) \sin^{-1}(\sin(i) / n)$ or $(r =) \sin^{-1}(\sin(60^\circ) / 1.6)$	C1
	33(°)	A1

9) 5054/21/M/J/17 Q10 — plane mirror, wavefront, reflection/refraction of waves, sound reflection — QP screenshot

- 10 (a) Fig. 10.1 shows two rays from an object that is placed in front of a plane mirror.

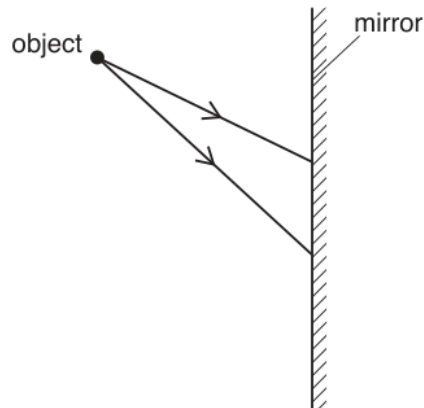


Fig. 10.1

- (i) On Fig. 10.1, draw the two reflected rays and locate the position of the image. [2]
- (ii) Apart from its position, state one characteristic of the image.

.....[1]

- (b) Fig. 10.2 shows wavefronts in a ripple tank. They move in the direction of the arrow.

The wave hits the boundary between two regions and the wave slows down as it enters the shaded region.

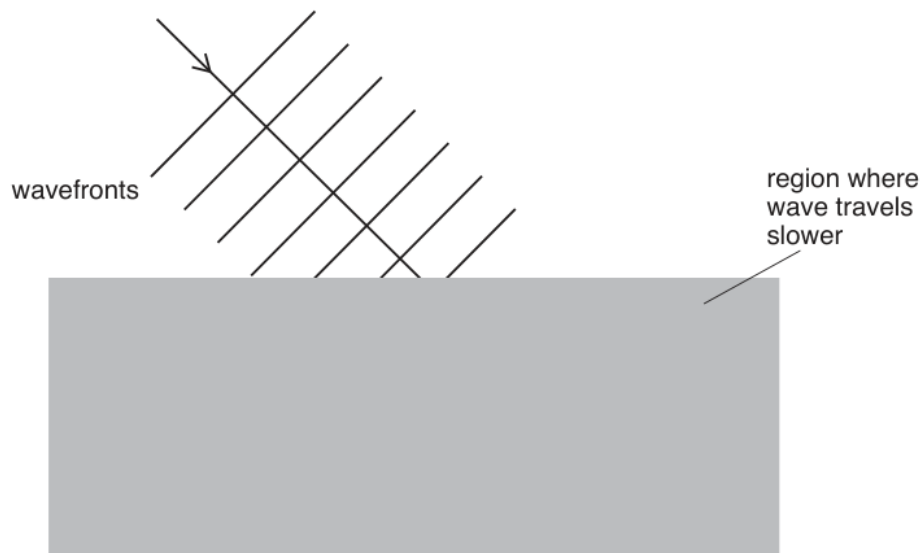


Fig. 10.2

- (i) State what is meant by *wavefront*.

.....

.....[2]

9) 5054/21/M/J/17 Q10 — plane mirror, wavefront, reflection/refraction of waves, sound reflection — QP screenshot

(ii) Both reflection and refraction occur at the boundary.

1. On Fig. 10.2, draw the reflected wavefronts. [1]

2. On Fig. 10.2, draw the refracted wavefronts. [2]

(c) A loudspeaker produces a sound of frequency 2.0 kHz. The wavelength of this sound in air is 16 cm.

(i) Calculate the speed of sound in air.

speed = [2]

(ii) 1. State the range of frequencies that can be heard by a healthy human ear.

..... [1]

2. Calculate the smallest wavelength of sound that can be heard by a healthy human ear.

wavelength = [1]

(iii) Describe a simple experiment to show that sound waves obey the law of reflection. You may draw a diagram if you wish.

.....

 [3]

9) 5054/21/M/J/17 Q10 — plane mirror, wavefront, reflection/refraction of waves, sound reflection — matching MS screenshot

5054/21

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PUBLISHED

May/June 2017

Question	Answer	Marks
10(a)(i)	both reflected rays correct by eye	B1
	image in correct position shown by continuation of rays behind mirror	B1
10(a)(ii)	virtual or upright or laterally inverted or same size (as object)	B1
10(b)(i)	line (joining points)	C1
	line joining points / particles on wave with same phase or line joining points along a crest etc.	A1
10(b)(ii)1	correct angle to surface $\pm 7^\circ$ with correct orientation and similar wavelength (by eye)	B1
10(b)(ii)2	at least two lines at smaller angle to surface with correct orientation in slower medium	B1
	showing smaller and constant wavelength with wavefronts deviated in correct direction	B1
10(c)(i)	$(v =) f\lambda$ or $2(000) \times 16$	C1
	32 000 cm / s or 320 m / s	A1
10(c)(ii)1	a range with 15–25 Hz as the lowest frequency and 15–30 kHz as the highest	B1
10(c)(ii)2	1.6 cm (if highest frequency is 20 000 Hz)	B1
10(c)(iii)	tube or other method to produce (narrow) beam of sound on source and/or on detector	B1
	stated detector or stated reflector	B1
	detector moved to find maximum loudness or angles i and r measured with suitable experiment	B1

9) 5054/21/M/J/17 Q10 — plane mirror, wavefront, reflection/refraction of waves, sound reflection — matching MS screenshot

5054/21

Cambridge O Level – Mark Scheme
PUBLISHED

May/June 2017

Question	Answer	Marks
11(a)	battery and ammeter symbols correct	B1
	thermistor symbol correct	B1
	all connected in series	B1
11(b)(i)1	0.05 (A) seen	C1
	$(V =) IR$ or 0.05×240	C1
	12 V	A1
11(b)(i)2	$(R =) V / I$ or 0.12 (A) seen	C1
	100 Ω	A1
11(b)(ii)	description of change in resistance in two equal changes in temperature, e.g. 240–100 and 100–48	C1
	difference values obtained are not the same, with one value shown e.g. 140 and 52 (Ω)	A1
11(c)(i)	two different metals connected together	B1
	voltmeter, ammeter or galvanometer in series	B1
	clear junction of two different metals used as the place to measure temperature	B1
11(c)(ii)	ANY 2 lines from measures high(er) temperatures more sensitive to changes in temperature measures fast(er) changing temperatures / reading can be taken more rapidly produce electronic output / output may be stored or processed / may be read remotely	B2